# Functional Forecasting of Turkish Electricity Load Curves

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Thesis Proposal

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# The Daily Load Curve: A Portrait of Economic Life

**Why the curve has the shape it has:**

- **Hours 0–5 (trough):** Most people asleep. Base-load plants (nuclear, hydro) dominate. Demand is low and very predictable.
- **Hours 6–8 (morning ramp):** Simultaneous awakening. Factories start shifts, HVAC switches on. Demand nearly **doubles in two hours**. Timing is sensitive to temperature and day type.
- **Hours 9–16 (plateau):** Industrial and commercial activity at full intensity. Predictable from calendar.
- **Hours 17–20 (evening surge):** People come home, cook, use AC or heating. Official forecast **most wrong here** (bias: 614–658 MWh/hour).
- **Hours 21–23 (wind-down):** Declining toward the trough.

### The forecasting object

$Y_d(h)$ = Turkish electricity load on day $d$, hour $h = 0, \ldots, 23$

Not 24 separate quantities — a single daily **curve**.

# Who Uses These Forecasts and Why Accuracy Matters

**The Turkish electricity market (post-2013 liberalisation):**

- **TEİAŞ** (TSO): publishes the official day-ahead load forecast used to schedule grid operations and clear the day-ahead market (EPIAS). *This is our benchmark.*
- **Generators** bid into the day-ahead market based on expected demand. Better demand forecasts ⇒ more strategic bids.
- **Large industrial consumers** face real-time imbalance penalties when their consumption deviates from contracted amounts. Better forecasts reduce penalties.
- **Unit commitment:** TEİAŞ decides which power plants to start up (hours in advance, significant startup costs). A better forecast ⇒ fewer unnecessary startups.

## The systematic underprediction puzzle

| Year | Off. bias | Our bias |
|---|---|---|
| 2023 | +344 MWh | +176 MWh |
| 2024 | +585 MWh | +34 MWh |
| 2025 | +472 MWh | +98 MWh |
| Avg. | +467 MWh | +103 MWh |

The official forecast **always underestimates** actual demand. The gap worsened in 2024 (hot summer + demand growth). Our rolling-origin model adapts continuously and nearly eliminates the bias in 2024 (+34 MWh).

*Why does the official underpredict?* Likely: model staleness, growing demand, asymmetric institutional incentives (over-procurement is visible; real-time reserve costs are not).

# From 24 Numbers to a Curve: Why FDA?

**The standard approach:** fit 24 separate forecasting models, one per hour. Problems:

- Ignores the shape of the curve — the morning ramp is a *joint* feature of hours 6, 7, and 8, not an independent event.
- Cannot borrow information across hours (hour 6 today predicts hour 7 today, not just hour 6 tomorrow).
- Cannot easily model "shape features" like ramp steepness or peak timing.

**The functional approach:** treat each day as a single mathematical object — a curve $Y_d(\cdot)$.

- Preserves intraday shape information.
- Allows differentiation: $DY_d(h) = \frac{d}{dh} Y_d(h)$ measures how fast demand is *changing* at each hour.
- FPCA compresses 24 correlated observations to $K = 2$ uncorrelated scores that carry 97.3%

**Representation:** B-spline basis ($M = 12$ functions)

$$Y_d(h) \approx \sum_{m=1}^{12} c_{dm} \, B_m(h)$$

then FPCA:

$$Y_d(h) = \mu(h) + \sum_{k=1}^{K} \xi_{dk} \, \phi_k(h) + \varepsilon_d(h)$$

## Forecast reconstruction

Once we forecast the $K$ scores $\hat{\xi}_d$, the full 24-hour load curve is recovered as

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\Phi}(h)' \hat{\xi}_d$$

# FPCA: Compressing the Curve to Two Numbers

**Variance decomposition** (full training data):

| FPC | Load var. | Cum. | Meaning |
|---|---|---|---|
| 1 | 93.4% | 93.4% | Level |
| 2 | 3.9% | 97.3% | Shape |
| 3 | 1.1% | 98.4% | — |
| 4 | 1.0% | 99.4% | — |

$K = 2$ exceeds the 95% cumulative variance rule while keeping the score vector parsimonious.

**Stability check (script 15):** We fit FPCA on 2019–2021, 2020–2022, and 2022–2024 separately. All pairwise inner products $> 0.993$. *The eigenfunctions do not change* — static FPCA is empirically justified.

**Economic meaning of the two components:**

**FPC 1 ("Level", 93.4%):** Its eigenfunction is nearly flat across all 24 hours. A high score $\xi_{d1}$ means uniformly high consumption — a busy industrial day, cold winter, hot summer. *Think of it as: how much electricity was used today overall.*

**FPC 2 ("Shape", 3.9%):** Its eigenfunction is positive at morning and evening peaks, negative at midday. A high $\xi_{d2}$ means pronounced morning/evening peaks relative to midday — typical of residential-heavy days (weekends, holidays). A low $\xi_{d2}$ means a flatter profile — typical of summer when AC smooths the curve. *Think of it as: how "peaky" today's curve is.*

# The Score Forecasting Model

After FPCA, the forecasting problem is: *given the score history, predict $(\xi_{d1}, \xi_{d2})$ for tomorrow.*

**FPCA VAR(1,7)** — core score dynamics:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + u_d$$

- **Lag 7:** Monday demand resembles last Monday's. Weekly seasonality is the dominant pattern in electricity use.
- **Lag 1:** yesterday's level and shape adjust the weekly baseline for recent demand shifts.
- Estimated by OLS equation by equation.

**Augmented model:**

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

**Feature blocks in $z_d$** (all predetermined at date $d$):

- **Calendar:** weekend, annual sine/cosine, summer break
- **Holiday:** fixed holidays, Ramadan, Bayram (Eid), holiday eves, post-holiday, bridge days
- **Load shape:** yesterday's mean, peak, range, rolling 7- and 28-day averages
- **Derivative features:** slope statistics from $DY_{d-1}(h) = \frac{d}{dh} Y_{d-1}(h)$ (mean slope, volatility, min/max, morning ramp, evening ramp, end-of-day drop)

*Holiday and derivative features are the two most important additions. Without holiday indicators, the model does not beat the official forecast.*

# Evaluation Design: No Leakage, No Shortcuts

**Rolling-origin evaluation** (walk-forward cross-validation for time series):

1. Test period: Jan 1 2023 – Dec 31 2025 (1,096 days)
2. For each test day $d$:
   - Estimate FPCA on all data before $d$
   - Estimate VAR + features on all data before $d$
   - Produce one-day-ahead forecast for $d$
   - Record error
3. Total: 26,304 hourly forecast errors evaluated

At every step, **only information dated $< d$ is used**. This mimics real operational conditions.

**Metrics:**

$$\text{MAE} = \frac{1}{N}\sum_{d,h} |e_{dh}|, \quad \text{MAPE} = \frac{1}{N}\sum_{d,h} \frac{|e_{dh}|}{Y_d(h)}$$

- **MAE:** absolute error in MWh; directly interpretable
- **MAPE:** percentage error; scale-free; standard in the electricity forecasting literature
- **RMSE:** penalises large errors more

**Statistical testing:** Diebold-Mariano test ($H_1$: challenger has lower expected absolute error than official). Applied pooled (all 26,304 observations) and hour by hour (24 separate tests).

*Comparing MAE numbers without a significance test is not enough.*

# Main Results

| Model | MAE | MAPE | MAE gain | MAPE gain | DM |
|---|---|---|---|---|---|
| Seasonal naive $Y_{d-7}$ | 1912 | 5.13% | −57.6% | −62.9% | — |
| Official forecast | 1214 | 3.15% | 0% | 0% | ref. |
| FPCA VAR(1,7) scores | 1443 | 3.76% | −18.9% | −19.4% | × |
| + ramp/calendar | 1238 | 3.26% | −2.1% | −3.7% | × |
| + holiday/load-shape | 1001 | 2.63% | +17.5% | +16.5% | *** |
| + derivative features | **980** | **2.56%** | +19.2% | +18.7% | *** |
| + residual step | 653 | 1.70% | +46.2% | +46.0% | *** |

DM: Diebold-Mariano test vs. official forecast (pooled hourly, $H_1$: challenger better). *** $p < 0.001$; $\times$ = not significant or worse than official.

- **Key insight:** the pure VAR dynamics are not enough. Holiday and load-shape features are what make the model competitive.
- **MAPE and MAE tell identical stories** — the gains are not a scale artifact.
- Main model beats official by **19% MAE** and **19% MAPE**, with $p \approx 10^{-163}$.

# Where the Model Wins and Where It Loses

**MAPE gain by hour (main model vs. official):**

| Hours | Regime | MAPE gain |
|---|---|---|
| 0–5 | Overnight trough | +14% to +60% |
| 6–8 | Morning ramp | −6% to −24% |
| 9 | Ramp end | ≈ 0% |
| 10–16 | Daytime plateau | +20% to +35% |
| 17 | Evening transition | +7% |
| 18–19 | Evening surge | −7% to −18% |
| 20–23 | Night wind-down | +10% to +20% |

## The ramp-hour puzzle

Our model **loses** at hours 6–8 (morning ramp) and 18–19 (evening surge) — the steepest transition hours.

**Why?** The official forecast incorporates weather (temperature) forecasts that determine the *timing* of ramps. Our model uses only historical load patterns.

A one-hour error in ramp timing propagates across the entire ramp — the slope error is not local.

DM test: model beats official at **19 of 24 hours**, statistically significant at **18 of 24 hours** (5% level). The residual step wins at **24 of 24 hours**.

# Diagnostics: Why the Residual Step Is Motivated

**ACF of forecast errors (Ljung-Box, lag 14):**

| Series | LB statistic | Conclusion |
|---|---|---|
| Hour 0 errors | 3,124 | Autocorrelated |
| Hour 6 errors | 7,227 | Strongly autocorrelated |
| Hour 8 errors | 994 | Autocorrelated |
| All 24 hours | — | All reject $H_0$ |

**Interpretation:** errors at ramp hours (especially hour 6) are correlated across consecutive days. If the model misses the ramp timing on Monday, it tends to miss it on Tuesday too. *This is a predictable pattern.*

## The residual correction

For target day $d$, using only past data:

$$\hat{e}_s(h) = Y_s(h) - \hat{Y}_s^{base}(h), \quad s < d$$

Estimate a correction curve $\hat{r}_d(h)$ from recent residuals and apply:

$$\hat{Y}_d^{res}(h) = \hat{Y}_d^{base}(h) + \lambda_d \, \hat{r}_d(h)$$

$\lambda_d$ is chosen by rolling-origin cross-validation on pre-$d$ data only. **No leakage.**

**Result:** 46% MAE gain, 46% MAPE gain vs. official. Wins at all 24 of 24 hours including the 5 ramp hours where the base model lost.

# Validation and Robustness

## 1. Diebold-Mariano test (pooled):

| Model | DM $p$-value |
|---|---|
| FPCA VAR(1,7) only | $\approx 1$ (worse) |
| + ramp/calendar | 0.996 (worse) |
| + holiday/load-shape | $10^{-135}$ |
| + derivative features | $10^{-163}$ |
| + residual step | $\approx 0$ |

The gains are not sampling variation.

## 2. Eigenfunction stability:

- Fit FPCA on 2019–21, 2020–22, 2022–24 separately.
- Pairwise inner products: all $> 0.993$.
- Static FPCA is empirically justified — no need for dynamic FPCA.

## 3. $K = 4$ robustness:

| $K$ | Resid. | MAE | MAPE | MAE gain |
|---|---|---|---|---|
| 2 | No | 980 | 2.56% | 19.2% |
| 2 | Yes | 653 | 1.70% | 46.2% |
| 4 | No | 801 | — | 34.0% |
| 4 | Yes | **640** | — | **47.3%** |

$K = 2$ is the conservative baseline (parsimony + variance rule). $K = 4$ is a robustness check.

## 4. Random forest benchmark:

| Model | MAE | MAPE | MAE gain |
|---|---|---|---|
| FPCA-RF (fixed basis) | 1217 | 3.14% | −0.3% |
| FPCA-OLS (main model) | **980** | **2.56%** | +19.2% |

RF ties the official; OLS beats it by 19%. **The linear score equation is adequate** — no meaningful nonlinearity left on the table.

# Thesis Contribution

**The coherent story in one flow:**

1. Turkish daily electricity load is a **curve**, not 24 separate numbers. FDA preserves intraday shape.
2. FPCA compresses to **2 scores** (97.3% variance). Eigenfunctions are empirically stable.
3. Scores follow a **VAR(1,7) + features**. Holiday and load-shape features are essential — pure dynamics fail.
4. The model beats the official forecast by **19% MAE / 19% MAPE**, $p \approx 10^{-163}$.
5. It loses at **ramp hours** (6–8, 18–19) where the official has a weather-forecast advantage.
6. **Error ACF** shows remaining serial structure. The residual step exploits this: 46% MAE gain, 24/24 hours.

**Methodological contributions:**

- Functional representation + FPCA score forecasting applied to the Turkish grid
- Systematic feature engineering from functional derivative
- Rigorous rolling-origin design with DM testing
- Empirical test of eigenfunction stability

**Economic contributions:**

- Document and explain the official forecast's systematic underprediction
- Identify ramp hours as the forecasting frontier (weather vs. historical patterns)
- Quantify the value of holiday and shape information for grid operators and market participants

# Selected References


Kokoszka, P. and Reimherr, M. (2017). *Introduction to Functional Data Analysis*. CRC Press.

Diebold, F.X. and Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.

Hyndman, R.J. and Ullah, M.S. (2007). Robust forecasting of mortality and fertility rates. *Computational Statistics & Data Analysis*, 51(10), 4942–4956.

Xie, J. and Hong, T. (2016). GEFCom2014 probabilistic electric load forecasting. *International Journal of Forecasting*, 32(3), 1012–1016.

Harvey, D., Leybourne, S. and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291.